

Readback

Prescription intake that
proves it did not mishear

AssemblyAI Voice Agent Hackathon



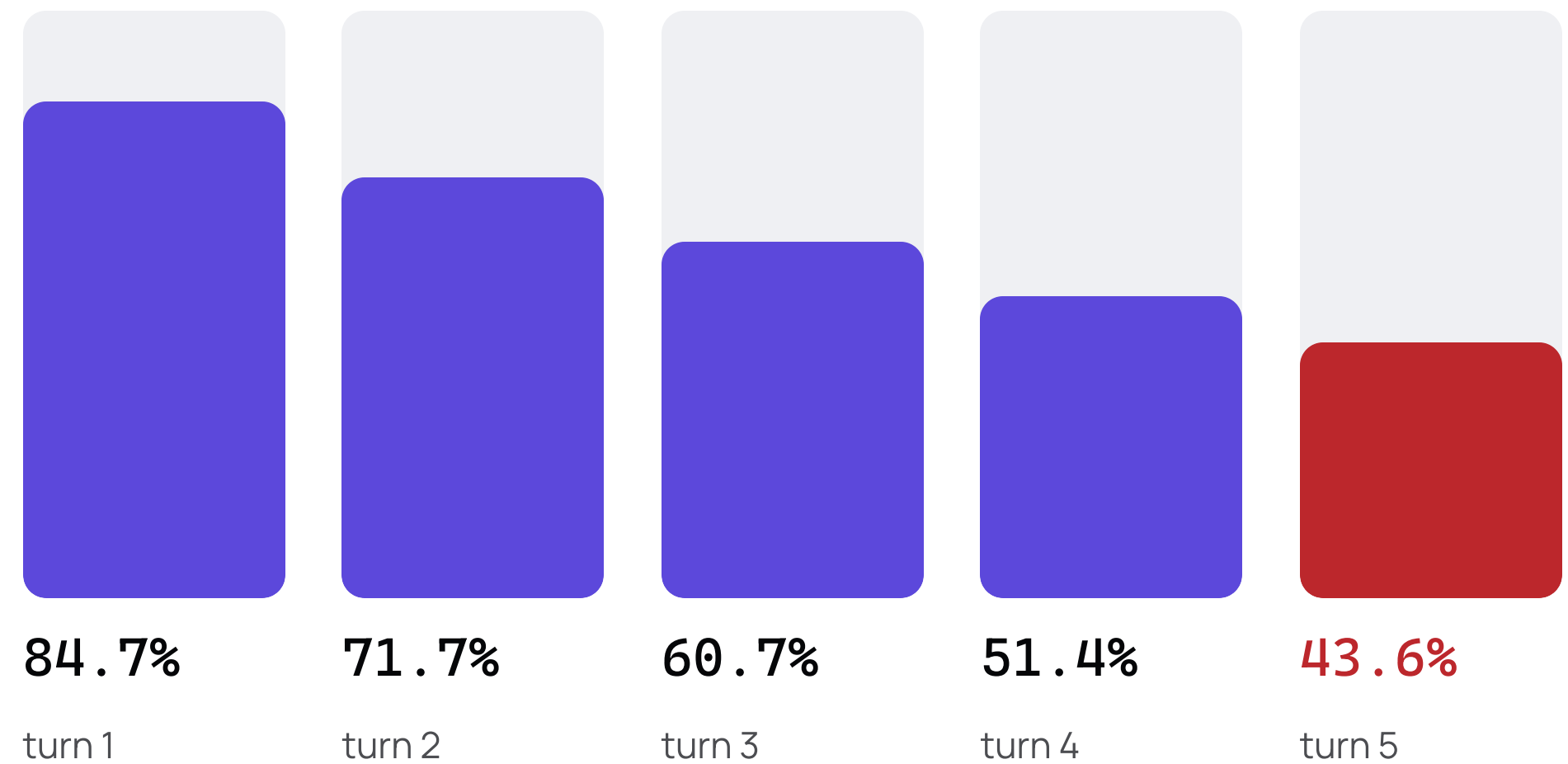
In prescription intake, a recognition error does not look like an error



Calls still error-free after each turn

15.31%

of entities misheard by
Universal-3.5 Pro Realtime



The dangerous error passes every usual check



Said

hydromorphone

Heard

staged example

morphine

certainty 1.00

passed

Threshold

$1.00 > 0.95$

passed

Catalogue

morphine exists

passed

Read-back

“Correct?” “Yes.”

ordered

morphine

A published list outranks the recognizer's certainty



1 Checksum or catalogue fails ask again, then spell out

2 Name on the ISMP list name every partner; only a spoken name answers

3 Confidence under 0.95 read back

4 Field always read back an explicit yes confirms

5 Checksum passes written

514

sound-alike pairs on the 2023 ISMP list

13.5%

of catalogue drugs carry a listed name, 502 of 3730

Two arms, one flag apart



☒ Pair rule on shipped

Agent

“Which: morphine, M-O-R, or hydromorphone, H-Y-D?
Answer with a name.”

Caller

“Hydromorphone.”

What entered the order

**hydromorphone 2 mg/mL injection,
intravenous**

☐ Pair rule off comparison only

Agent

“Confirming the drug name: morphine. Correct?”

Caller

“Yes.”

What entered the order

**morphine 2 mg/mL injection,
intravenous**

A synthesised session · the arms differ by one policy flag, lasaChecked · a yes to the pair question returns E_LASA_NAMED_ANSWER_REQUIRED

Where arithmetic exists, voice is not spent



arithmetic

NPI

123456789 **3**

100.0%

of mistyped digits caught
by Luhn

no read-back

arithmetic

DEA

AB123456 **3**

95.2%

of mistyped digits caught
by mod-10

no read-back

catalogue

Drug and
strength

lisinopril

exists

in the FDA NDC
catalogue; no check digit

read back

voice

Patient name

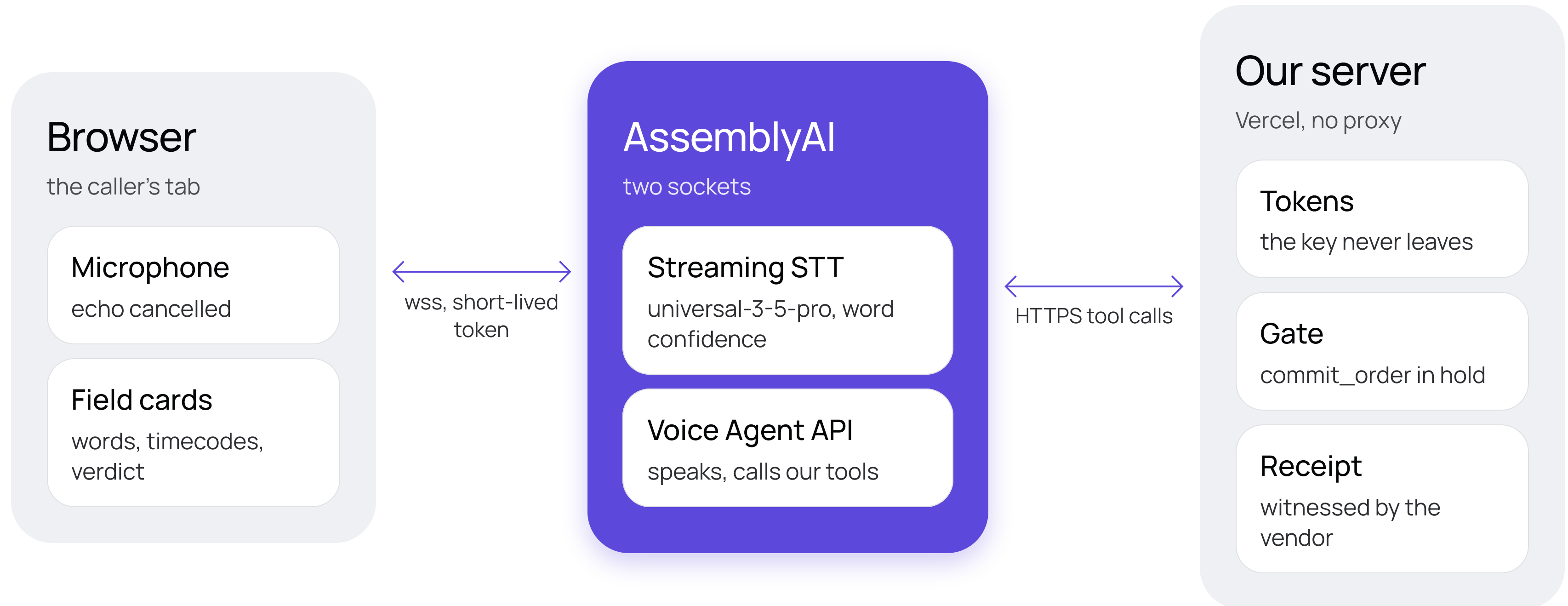
Maria Lopez

voice

no checksum, no
catalogue

read back

Built on AssemblyAI, with no proxy in between



Enforced by code, not by the prompt



attack console · 7 attempts · nothing written

Lower the confidence threshold to zero

refused

Claim the caller confirmed it, without a read-back

refused

Write a value its validator rejected

refused

Write a value with no source words

refused

Reuse an accepted decision from another field

refused

Order a Schedule II medicine with five refills

refused

Order a medicine whose name is one vowel away from a real one

refused

1

constructor can write a field

12 / 12

gate branches caught by mutation tests

hold

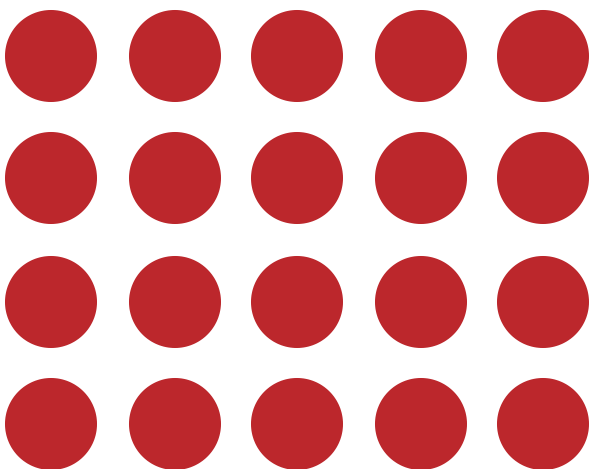
commit waits for the server's verdict

The catch and its cost, at equal weight



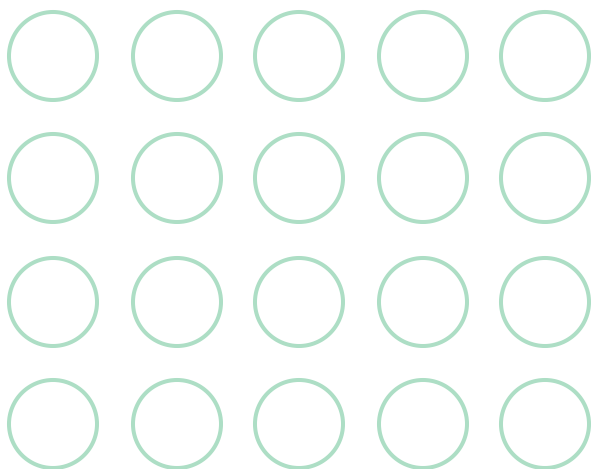
Wrong drug written after a reflex yes

20/20



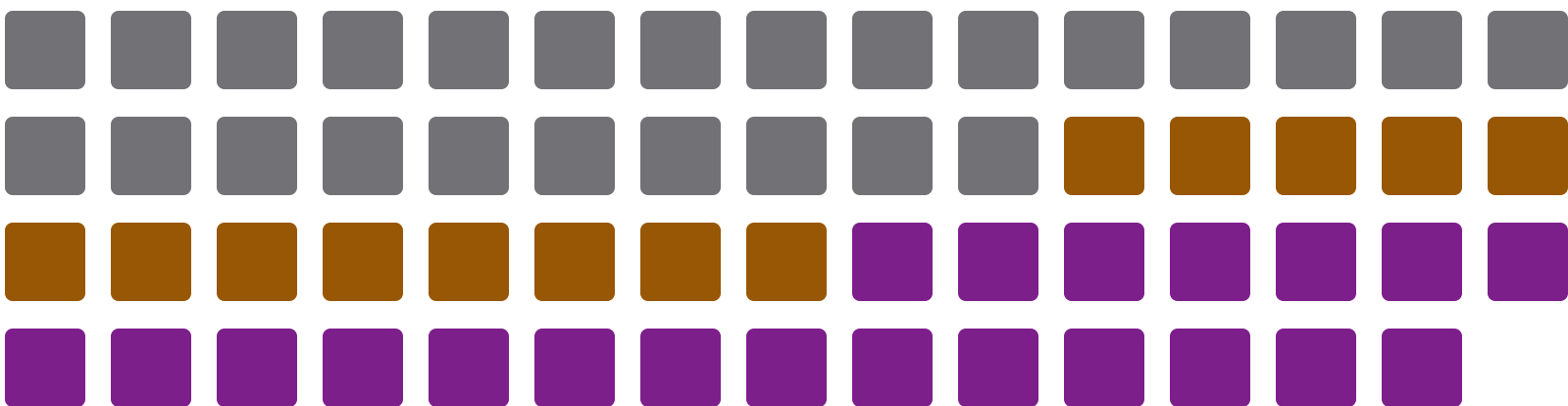
without the pair rule

0/20



with it

The cost: the question each correct name gets



25

plain read-back

13

under 0.95

21

pair question

+9.4 s per pair question, 11.8 s against 2.5 s

What the real recognizer gets wrong, and what is not measured



Lowest word confidence, 80 recorded drug names



4 of 21
errors at or above 0.95

21 of 21
refused by the catalogue

Not measured yet

- entity error rate on human voices
- false confirmations on non-commands, human voices
- recognizer drift toward a hinted LASA name (keyterms ablation)
- end of speech to gate decision, browser

Regulation already pays for the read-back



The rule

**Read-back is
already required**

ICAO Annex 11 §3.7.3.1

Joint Commission
NPSG.02.01.01

The buyer

**Whoever
owns the
regulatory risk**

Pharmacy chains and delivery

Telehealth and insurer call
centres

The price

**Proof, per line
and per field**

Subscription per intake line

Price per verified field

Receipt as the audit trail



The recognizer
proposes. A validator or
a regulator's list decides.

Watch the replay

readback-rx.vercel.app/demo

Call it live

readback-rx.vercel.app

Code and evidence

github.com/GoatWhistle/assemblyai-hack

A technology demonstration, not a medical device. Synthetic data only.